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Specifying and Analyzing Transactional Consistency Models with Predicates

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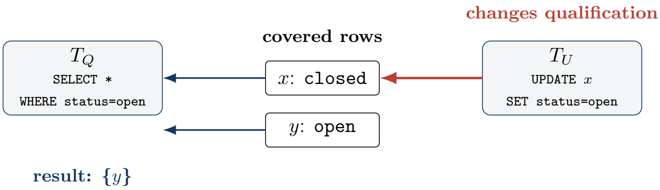
A semantic and graph-theoretic foundation for predicate operations

A predicate result contains more than returned rows



Key observation: $x \notin$ result
still records an observation of x.

Item reads: value observed

Predicate reads: returned rows **and excluded rows**

Phantom-style interactions need an explanation for both.

Berenson et al., 1995; Adya, 1999

The gap: familiar foundations stop at named items

Execution-level specifications $\rightarrow$ Dependency-graph characterizations $\rightarrow$ Checking / robustness

Predicate operations are missing from the semantic–graph correspondence.

Question: Can we recover an exact, predicate-aware foundation for SER and SI?

Semantic anchor: explain every covered item

VIS/AR execution framework

Visible transactions

provide the version set

$\Longrightarrow$

Latest visible writer

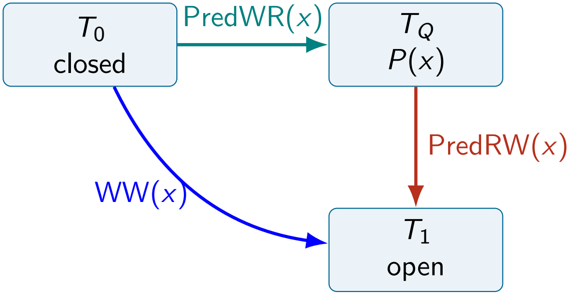
explains each covered item

Returned item: witness matches P

Omitted item: witness does not match P

Cerone, Bernardi, and Gotsman, 2015

From a semantic witness to graph edges



PredWR: version witness

PredRW: later writer changes the observed result

Result-sensitive: $\Delta(T_0, T_Q, T_1, x)$

The graph records why the result was observed — including the absence of x.

Exact graph characterizations

Serializability

$$\mathcal{H} \models \mathsf{SER}$$

$$\iff$$

internally consistent $+$
acyclic predicate-aware graph

Snapshot isolation

$$\mathcal{H} \models \mathsf{SI}$$

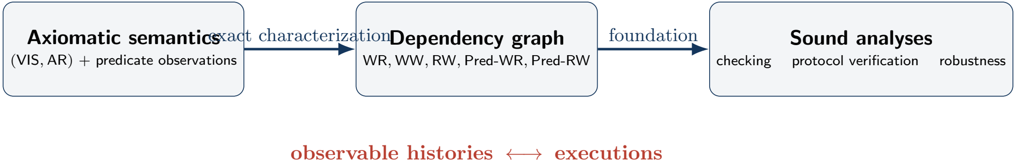
$$\iff$$

internally consistent $+$
every cycle has two adjacent
RW / PredRW edges

Not analogy: both equivalences are proved against the predicate-aware axioms.

Cerone and Gotsman, 2018

Why the proof bridge matters



- Predicate graphs are **observable-history** objects.
- Axiomatics supplies the **semantic** correctness target.
- The bridge validates graph-based analysis beyond item-only workloads.

Related work disagrees on one crucial edge

Pred-RW^{-}
earliest match-changing writer

$\xrightarrow{\subseteq}$

Pred-RW
any match-changing writer

$\xrightarrow{\subseteq}$

Pred-RW^{+}
every later writer

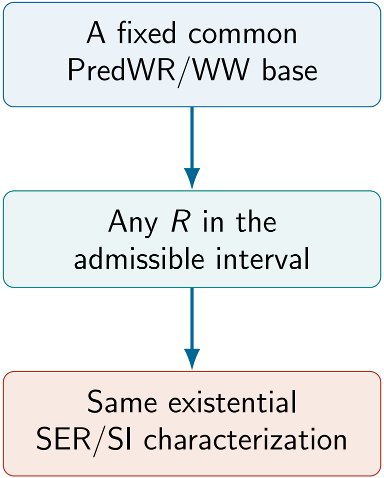
Same existential history-level characterization of SER and SI
edge sets and checking cost may differ

Sun and Zou, 2025

Adya, 1999; “Generalized Isolation Level Definitions” 2000

Bouajjani, Enea, and Román-Calvo, 2025

What the design space tells us



The theorem is semantic, not a claim that edge sets coincide.

Practical consequence: choose an admissible edge policy for an analysis, while retaining its history-level meaning.

Checking cost can still differ.

Future work: turn the foundation into analysis tools

Efficient black-box checking

Recover version sets and version order

Robustness analysis

Extend static reasoning to predicate operations

Both need a semantics-preserving predicate dependency graph.

References I

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Take away

Predicate results are observations over a covered set.

axiomatic predicate semantics → predicate-aware dependency graphs → exact SER/SI characterizations

A principled basis for checking and analyzing databases with predicates.